

MDSA20030

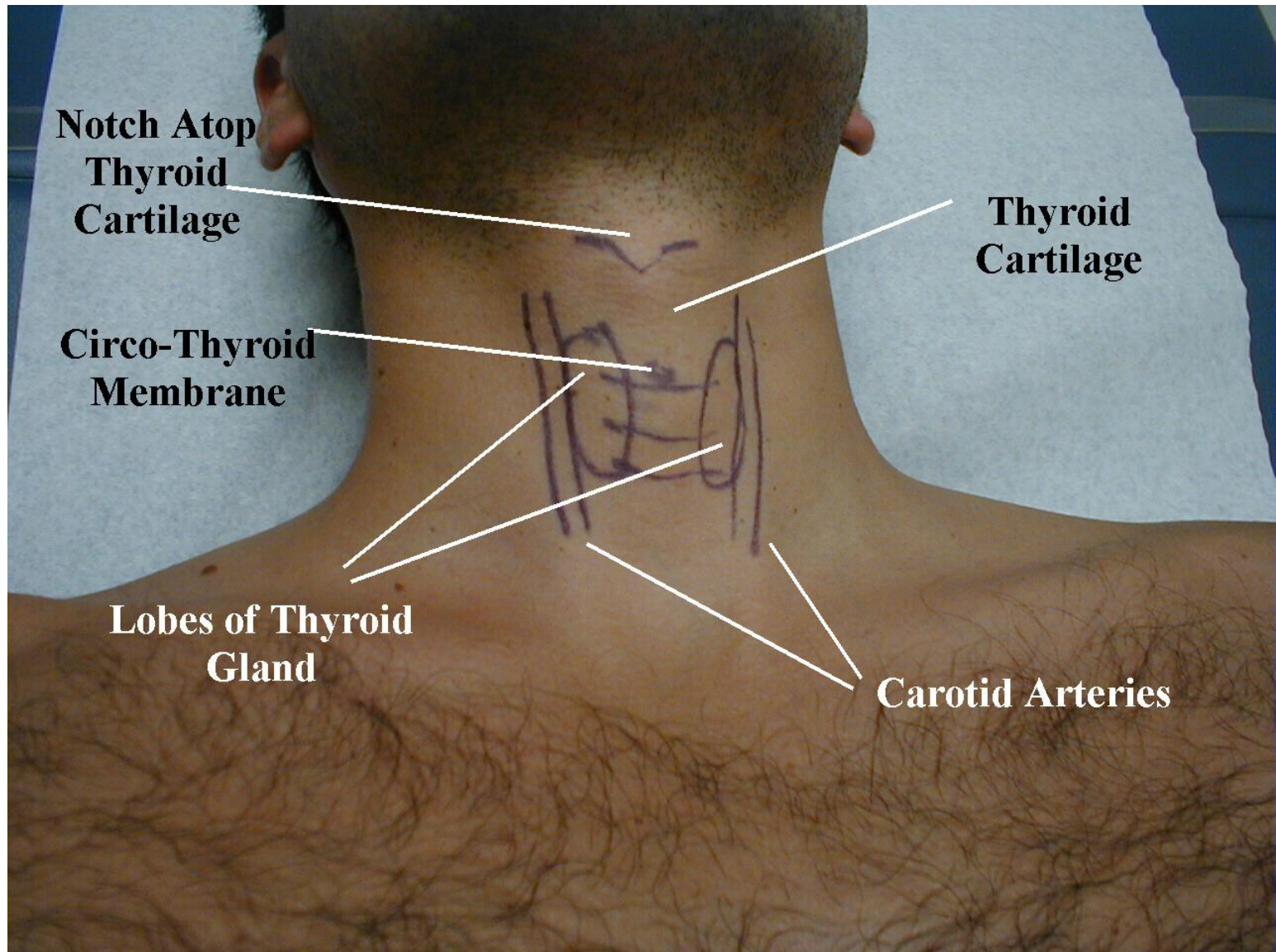
**Endocrine
Biology**

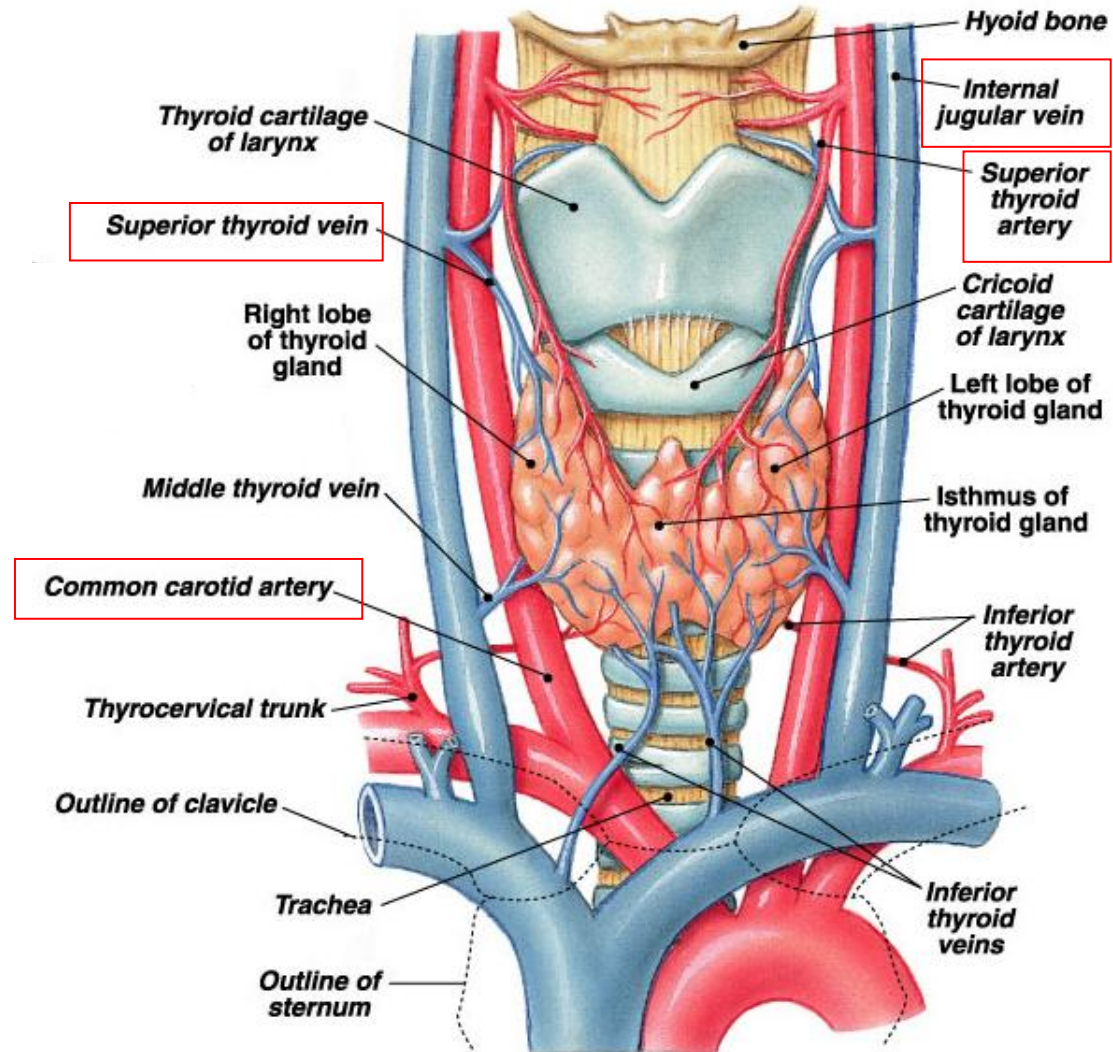
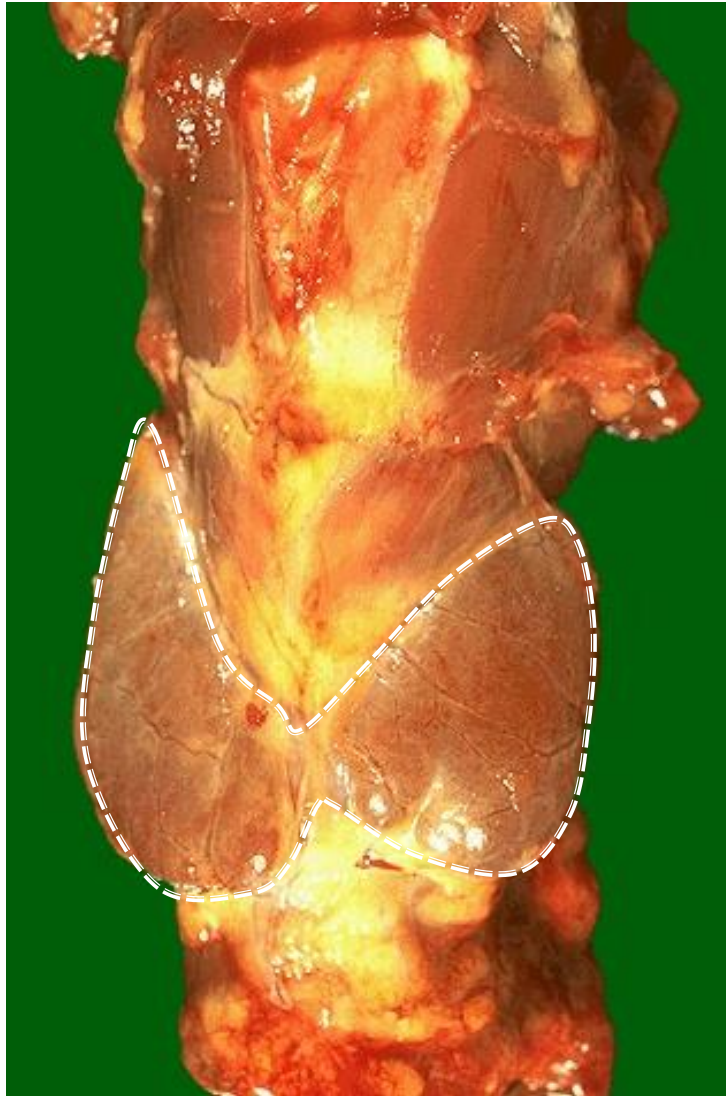
**Physiology of
the
Thyroid Gland**



Robert Graves (1796-1853)

Thyroid Anatomy





Thyroid Histology

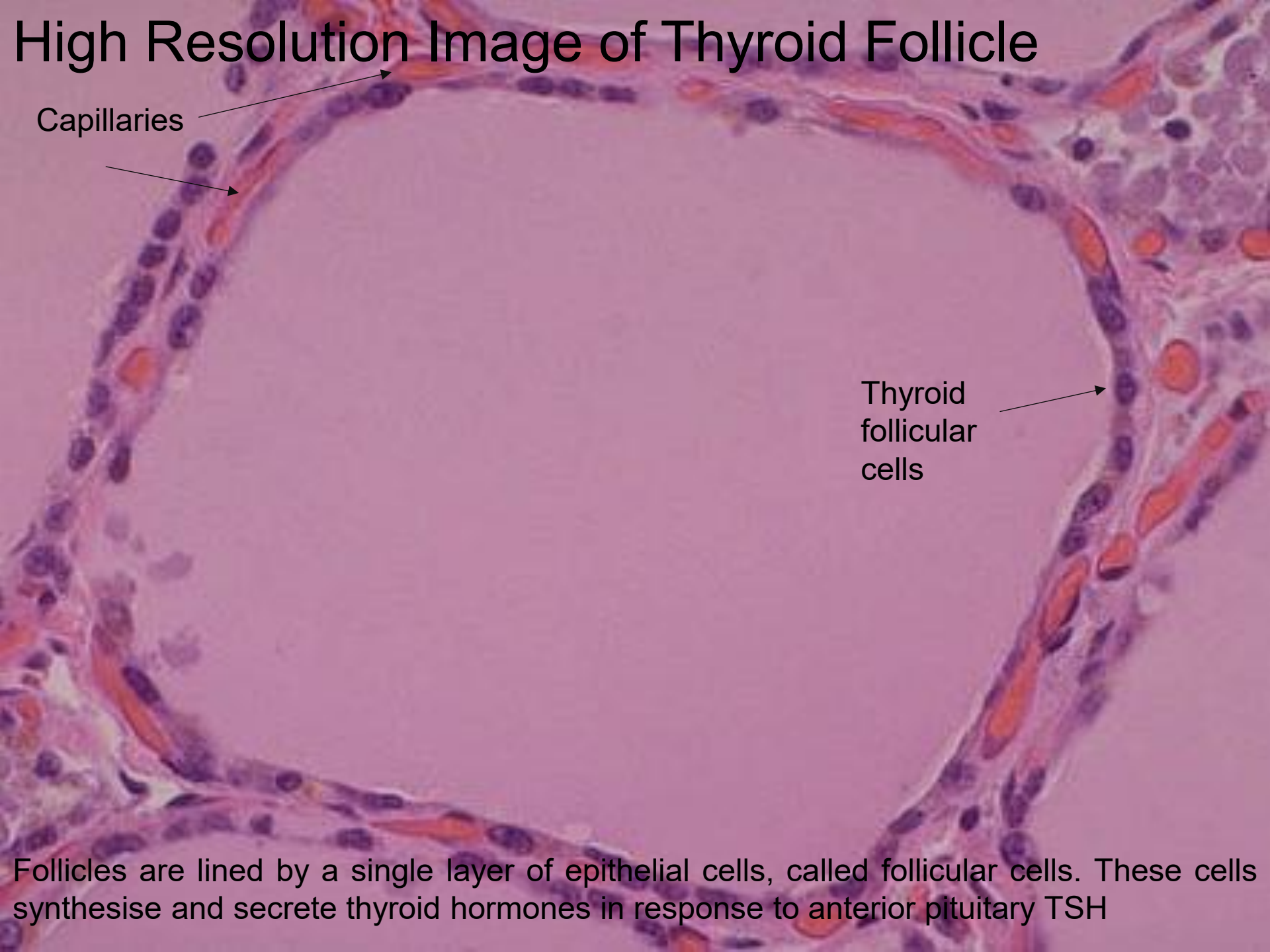


Note follicular structure. The follicles contain a proteinaceous material called colloid – this is essentially an extracellular storage form of thyroid hormone

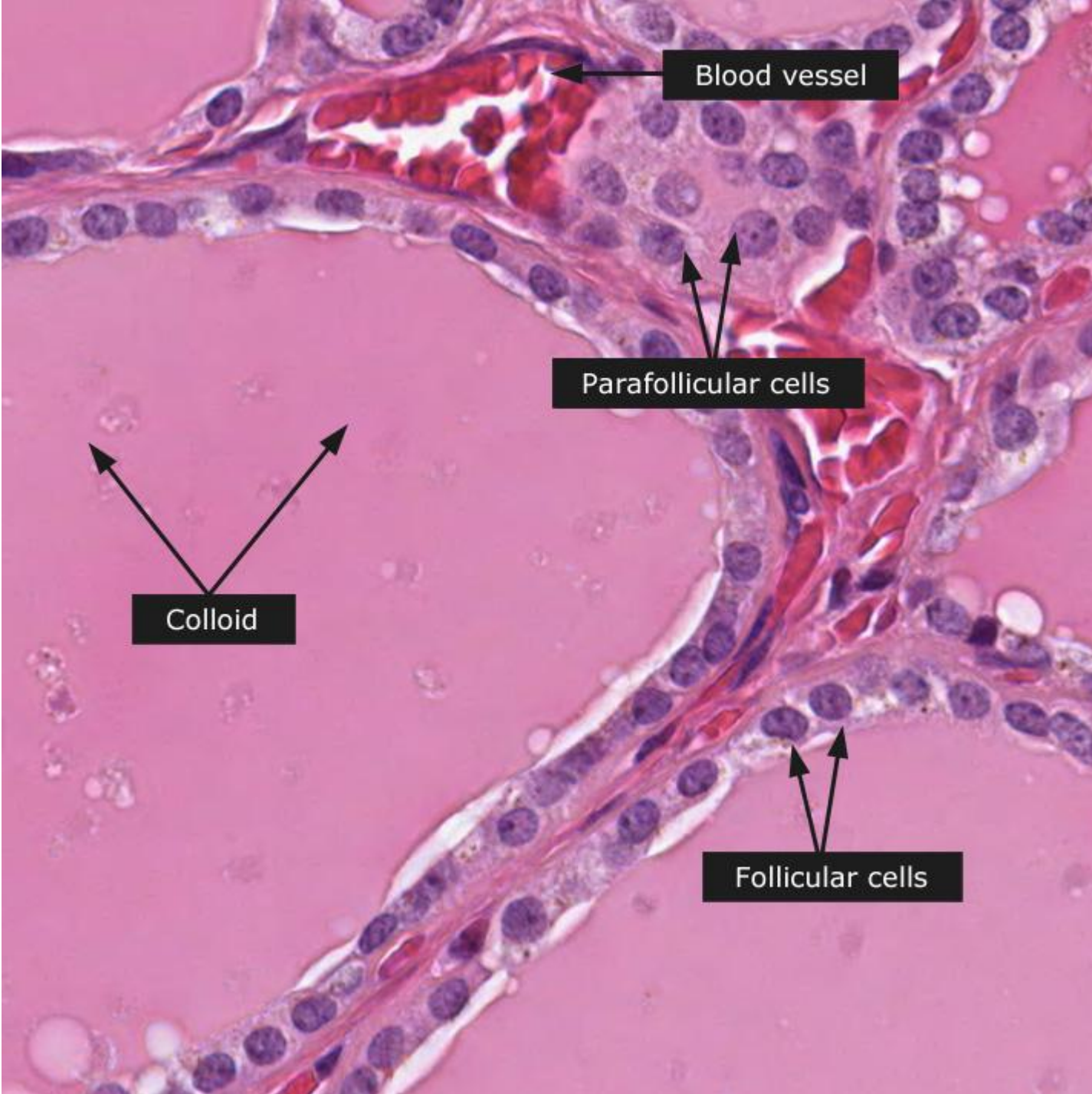
High Resolution Image of Thyroid Follicle

Capillaries

Thyroid
follicular
cells



Follicles are lined by a single layer of epithelial cells, called follicular cells. These cells synthesise and secrete thyroid hormones in response to anterior pituitary TSH

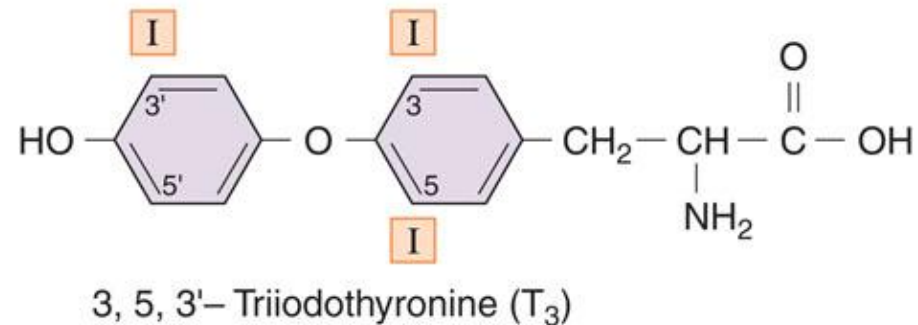
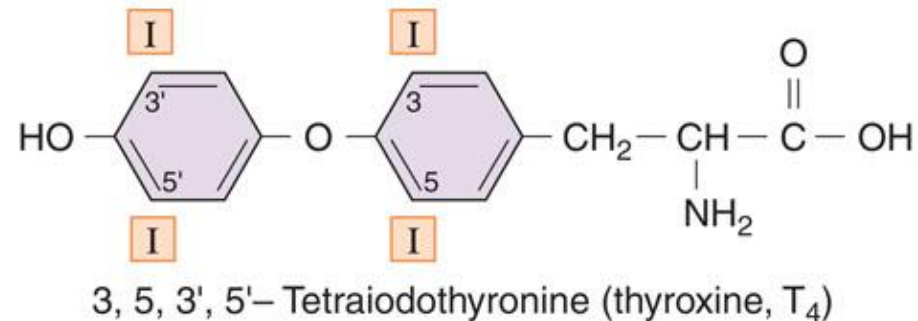
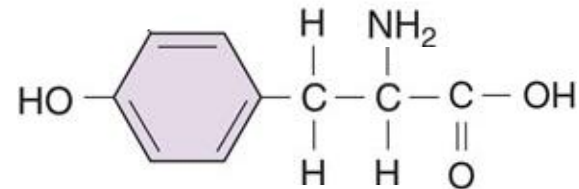


There is a second type of endocrine cell in the thyroid gland, usually found in the intrafollicular region. These are called parafollicular or C (clear) cells; they secrete a peptide hormone called calcitonin which plays a role in calcium homeostasis. Although it is produced in the thyroid gland, it is not considered a thyroid hormone

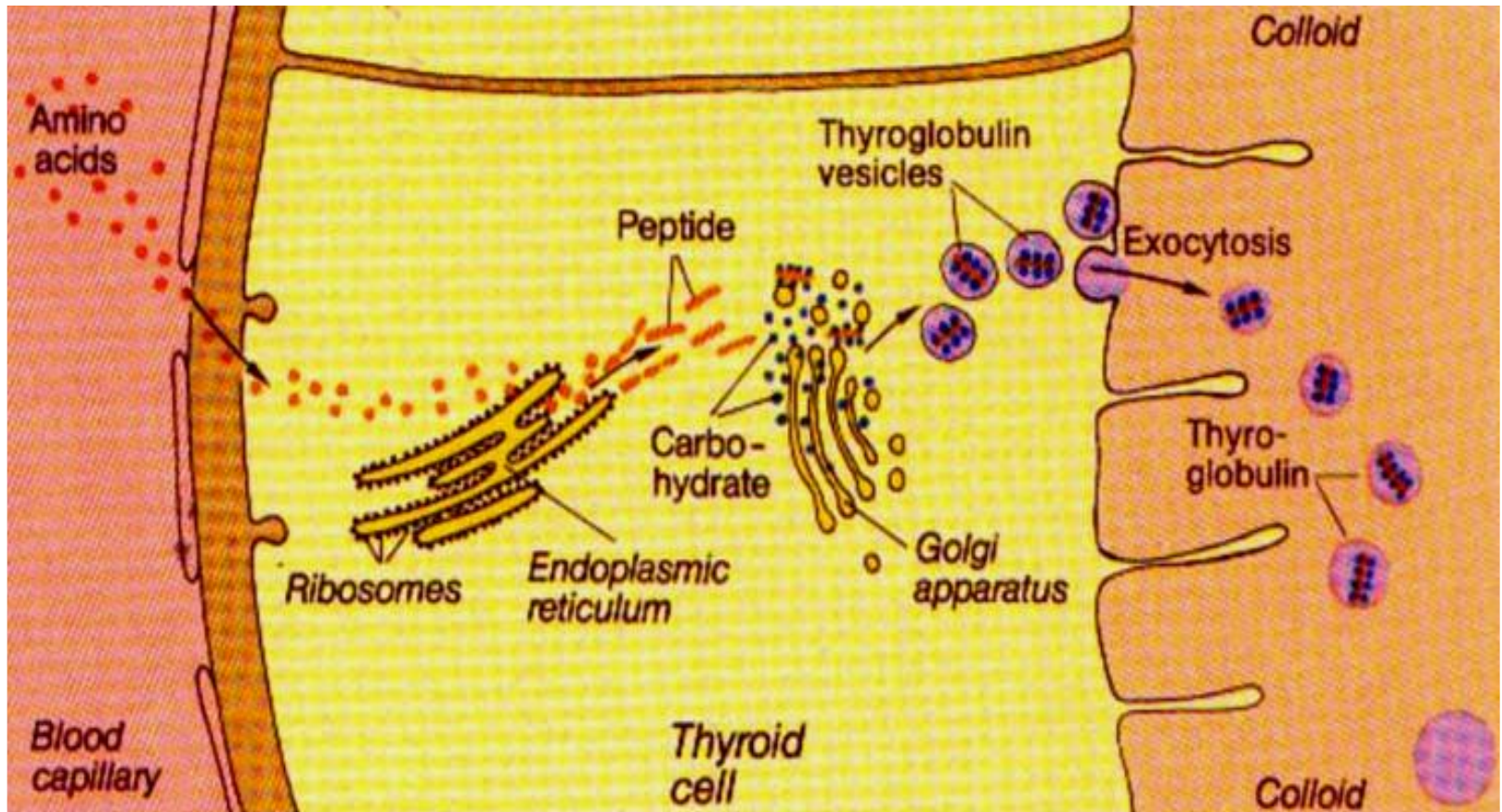
Thyroid Hormones – T3 and T4

- Synthesis requires iodine
 - ~ 150-200µg/day
- Iodination at C3 and C5 positions of outer (denoted by ') and inner rings
- Moderately non polar with limited solubility in plasma

tyrosine (tyr)

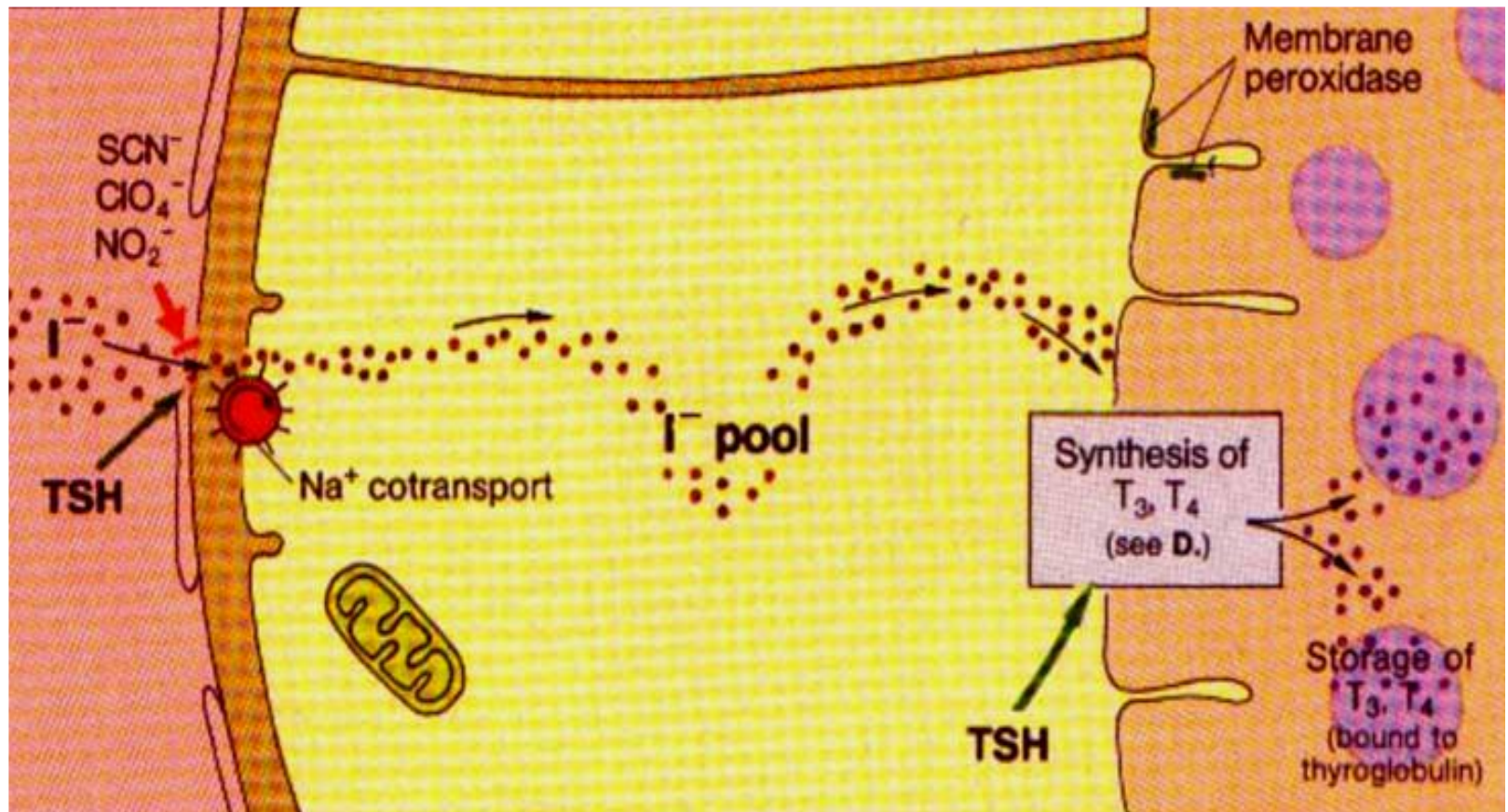


Thyroid Hormone Biosynthesis: Synthesis of Thyroglobulin

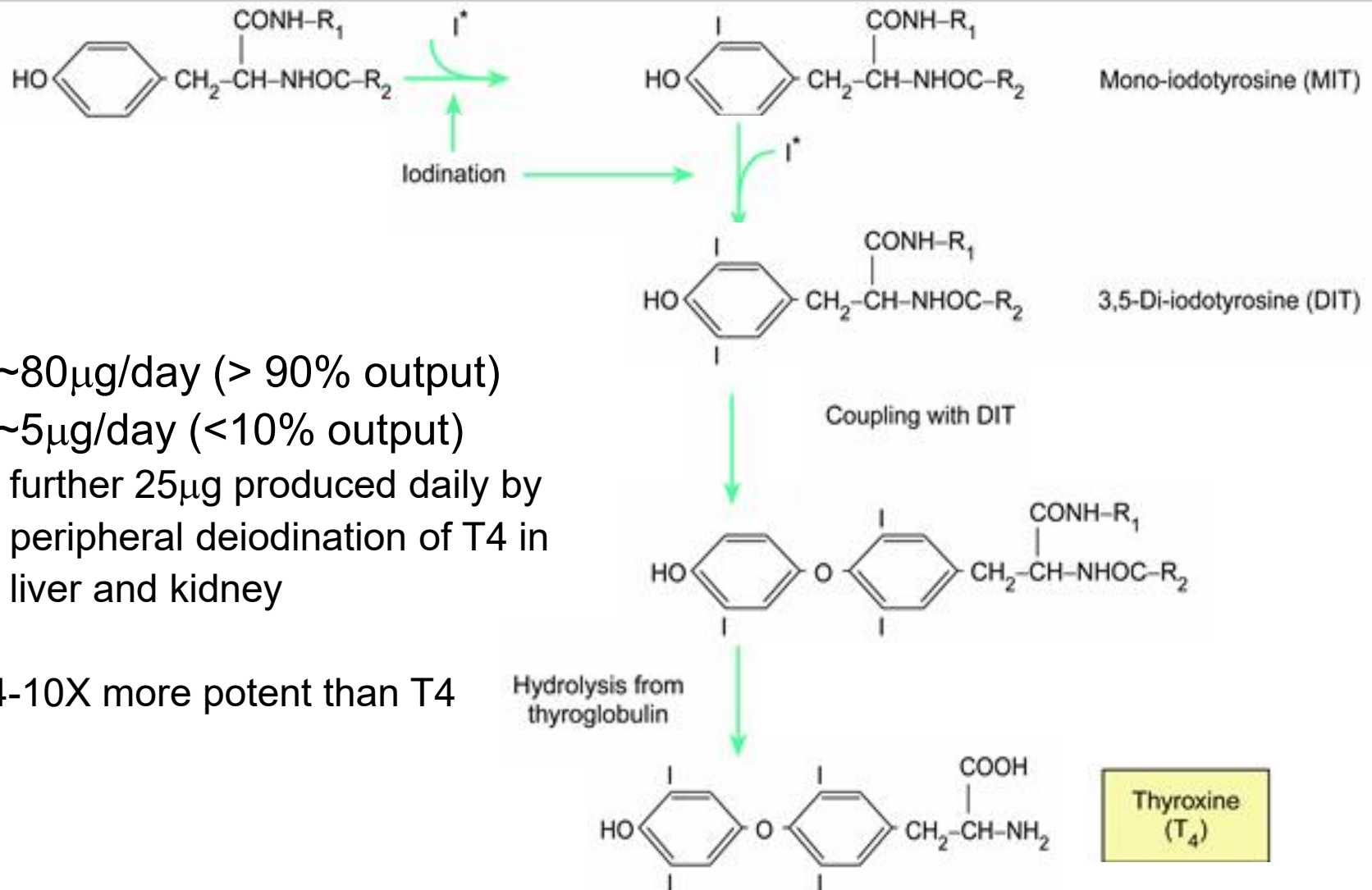


contains 140 tyrosine residues

Thyroid Hormone Biosynthesis: Iodide Trapping



TH Synthesis



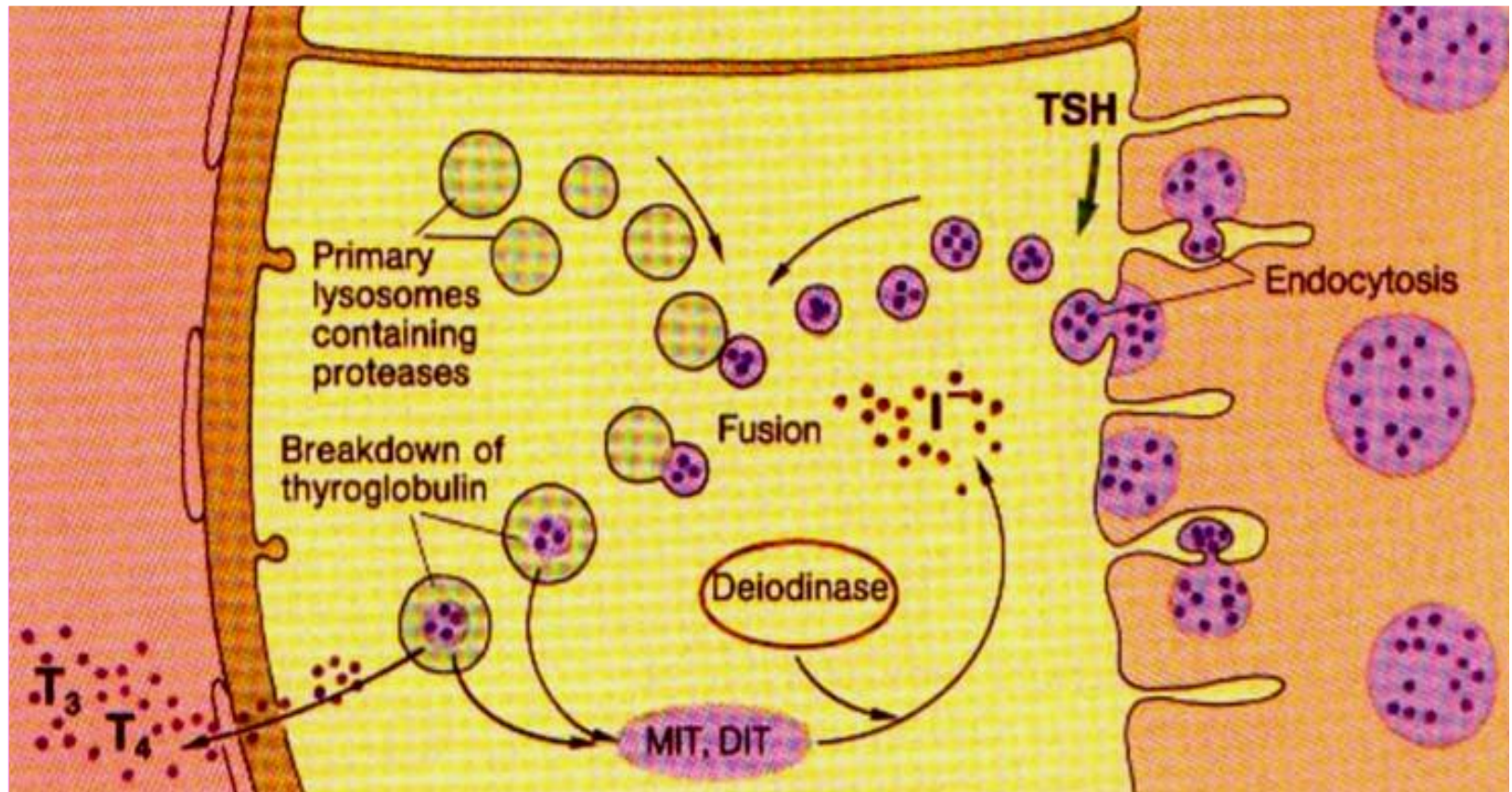
T4 $\sim 80\mu\text{g/day}$ ($> 90\%$ output)

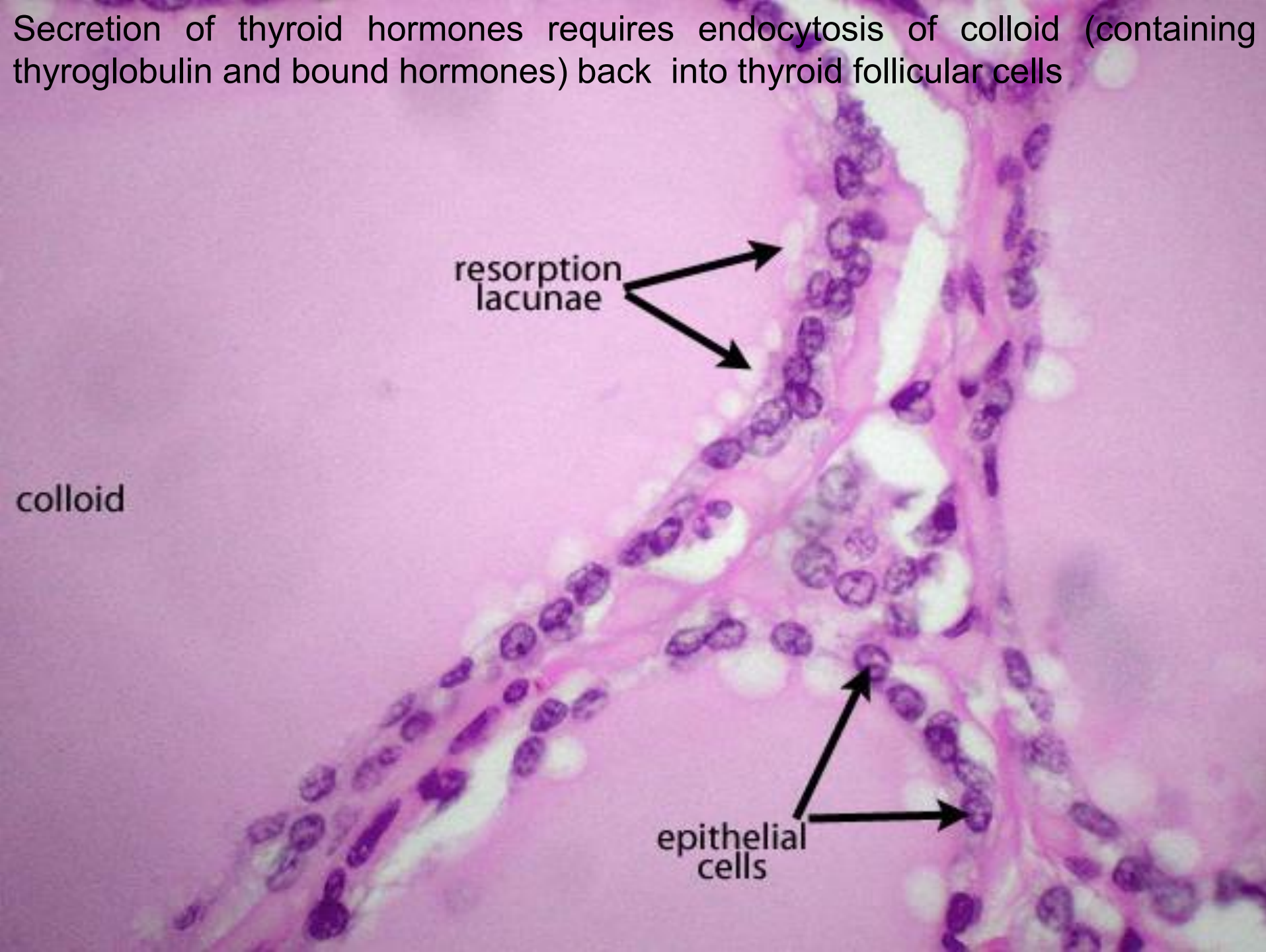
T3 $\sim 5\mu\text{g/day}$ ($< 10\%$ output)

further $25\mu\text{g}$ produced daily by
peripheral deiodination of T4 in
liver and kidney

T3 4-10X more potent than T4

Thyroid Hormone Biosynthesis: T3/T4 Secretion





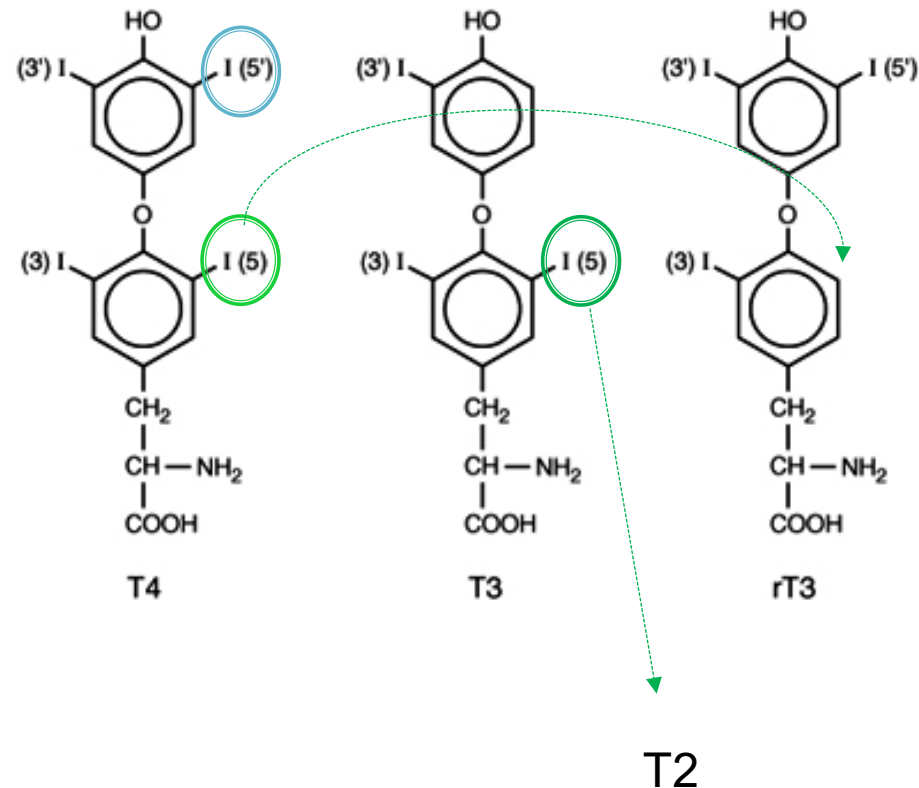
Plasma Transport of T3/T4

- Normal circulating concentrations T4 ~ 4.5-11 $\mu\text{g/dL}$, T3 ~ 60-180 ng/dL
- <0.5% exist as free hormone, remainder bound to 'carrier' proteins
- Major plasma transport proteins
 - thyroxine binding globulin (TBG)
 - higher affinity for T₄ than T₃
 - plasma $t_{1/2}$ T4 $t_{1/2}$ >>> T3
 - TBPA (transthyretin) intermediate binding affinity
 - albumin - lowest affinity but highest abundance
 - readily dissociable

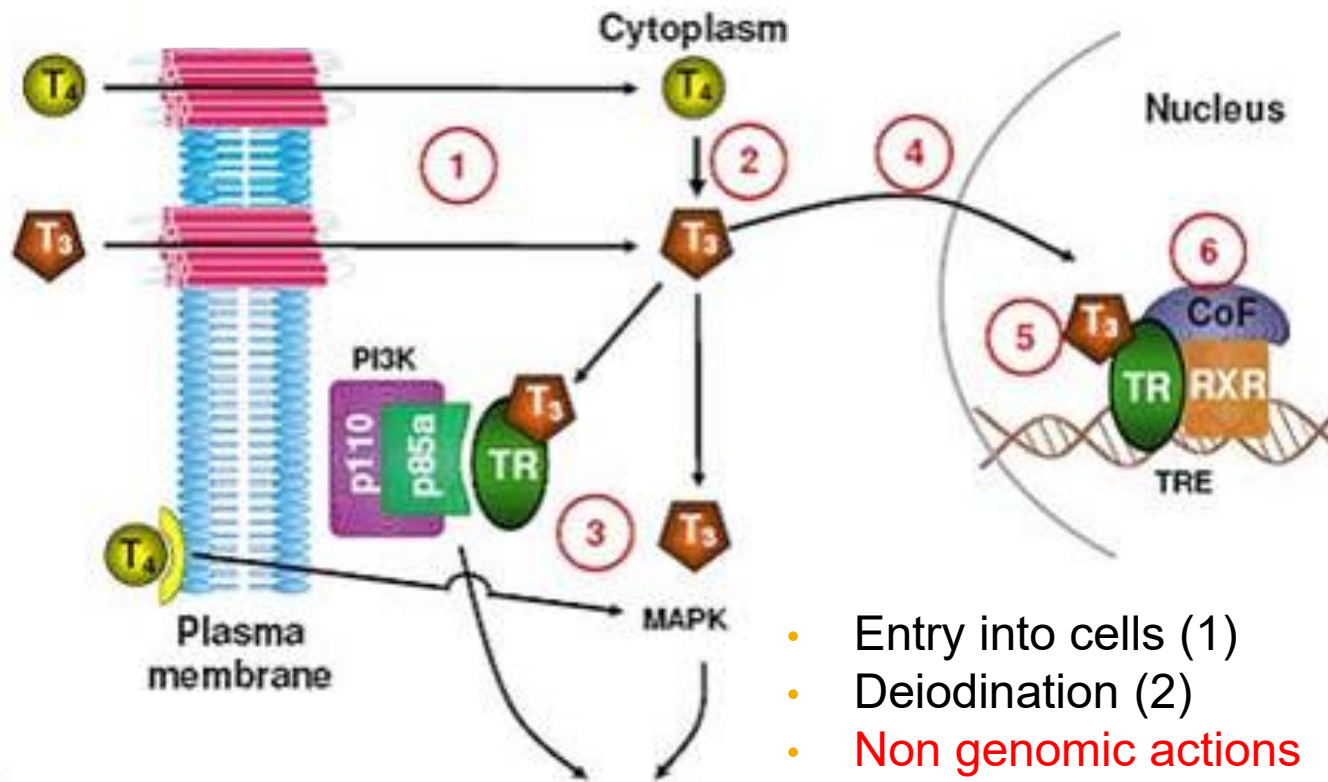
	TBG	TBPA	Alb
T4 (%)	68	11	20
T3 (%)	80	9	11
conc (mg/100ml)	2.0	25	4200

Thyroid Hormone Deiodination

- Deiodinases are selenium dependent enzymes
- Play a role in the activation and inactivation of THs
- **5' monodeiodinase** converts T4 to T3 in target cells (C5 outer ring)
 - enzyme less active during fasting / starvation and cortisol excess
- **5 monodeiodinase** inactivates T4 and T3 (removes C5 inner ring)
 - converts T4 to r(everse)T3 or T3 to T2



Thyroid Hormone Mechanism of Action



TRE

AGGTCATGACCT
TCCAGTACTGGA

- Entry into cells (1)
- Deiodination (2)
- **Non genomic actions (3)**
- Nuclear import (4)
- Binding to TH receptor, formation of heterodimer with retinoic acid receptor and binding to TRE (5)
- Recruitment of co-activator proteins (6)

Thyroid Hormones and Basal Metabolic Rate

- Metabolic rate (MR) – total rate of body metabolism
 - amount of O₂ consumed by body/min
- Basal metabolic rate (BMR) – MR of awake relaxed person 12–14 hrs after eating at room temperature
 - basic metabolic/energetic cost of life
 - ‘idling’ speed of the body
 - ~ 70% calories consumed expended on maintaining BMR
- BMR depends on age, sex, body surface area, activity level, body composition, ambient temp etc.
 - strongly influenced by thyroid hormone levels
 - hyperthyroidism: high BMR
 - hypothyroidism: low BMR

Metabolic Effects of T_3

- Stimulate aerobic metabolism
 - increase mitochondrial number and size
 - increase O_2 consumption
- Generally stimulates all aspects of carbohydrate metabolism and increase protein turnover
- Stimulates lipolysis and release of free fatty acids and glycerol
- Stimulates metabolism of cholesterol to bile acids
- Facilitates removal of LDL from plasma

Role in Growth and Development

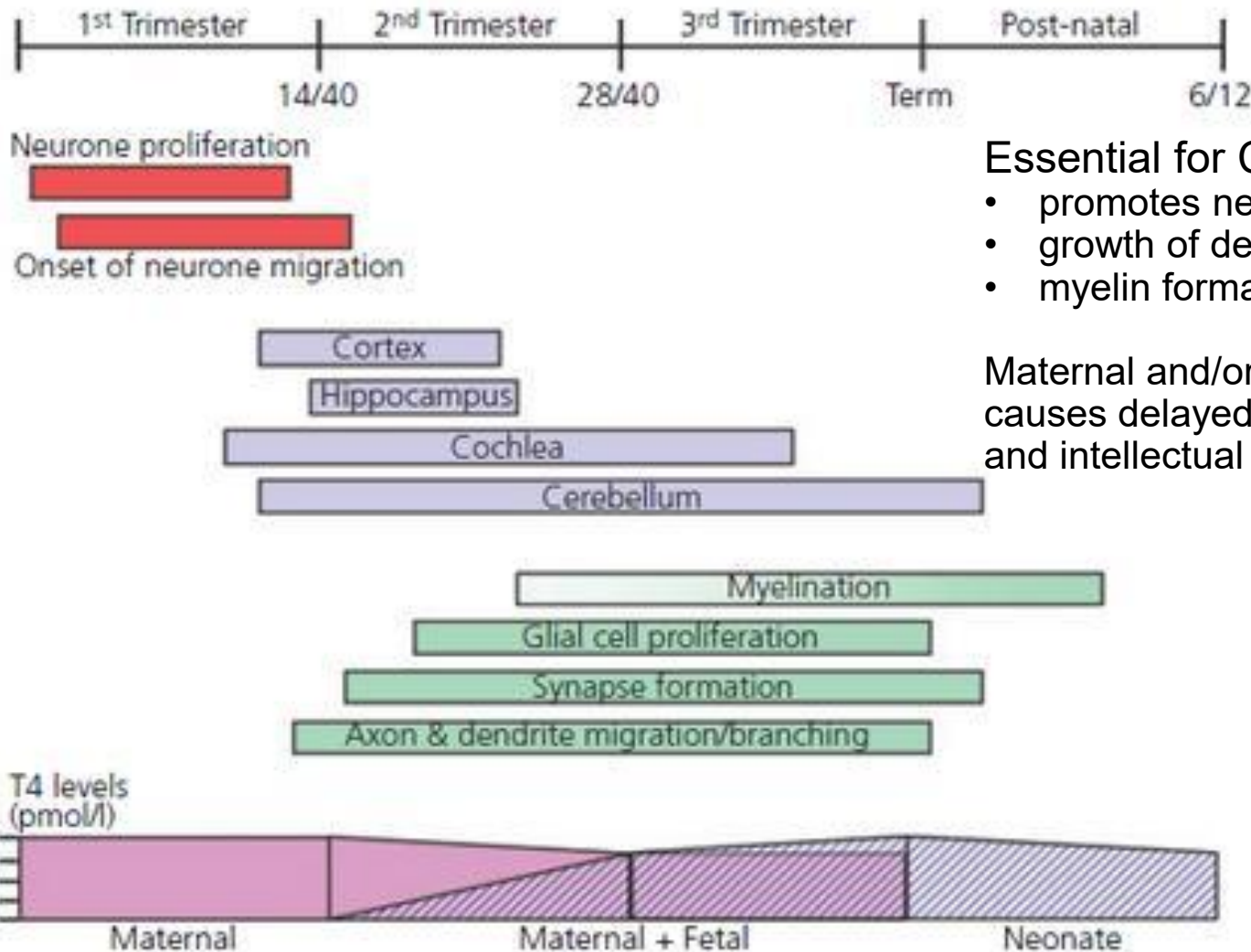
- Essential for normal growth
 - increases protein synthesis
 - increases energy production and substrate availability
 - permissive for anabolic actions of growth hormone
- Essential for CNS development
 - promotes nerve growth
 - promotes growth of dendrite cells
 - promotes myelin formation
 - cretinism
 - delayed CNS development and intellectual disability



age 6

age 17

Thyroid Hormone and Brain Development

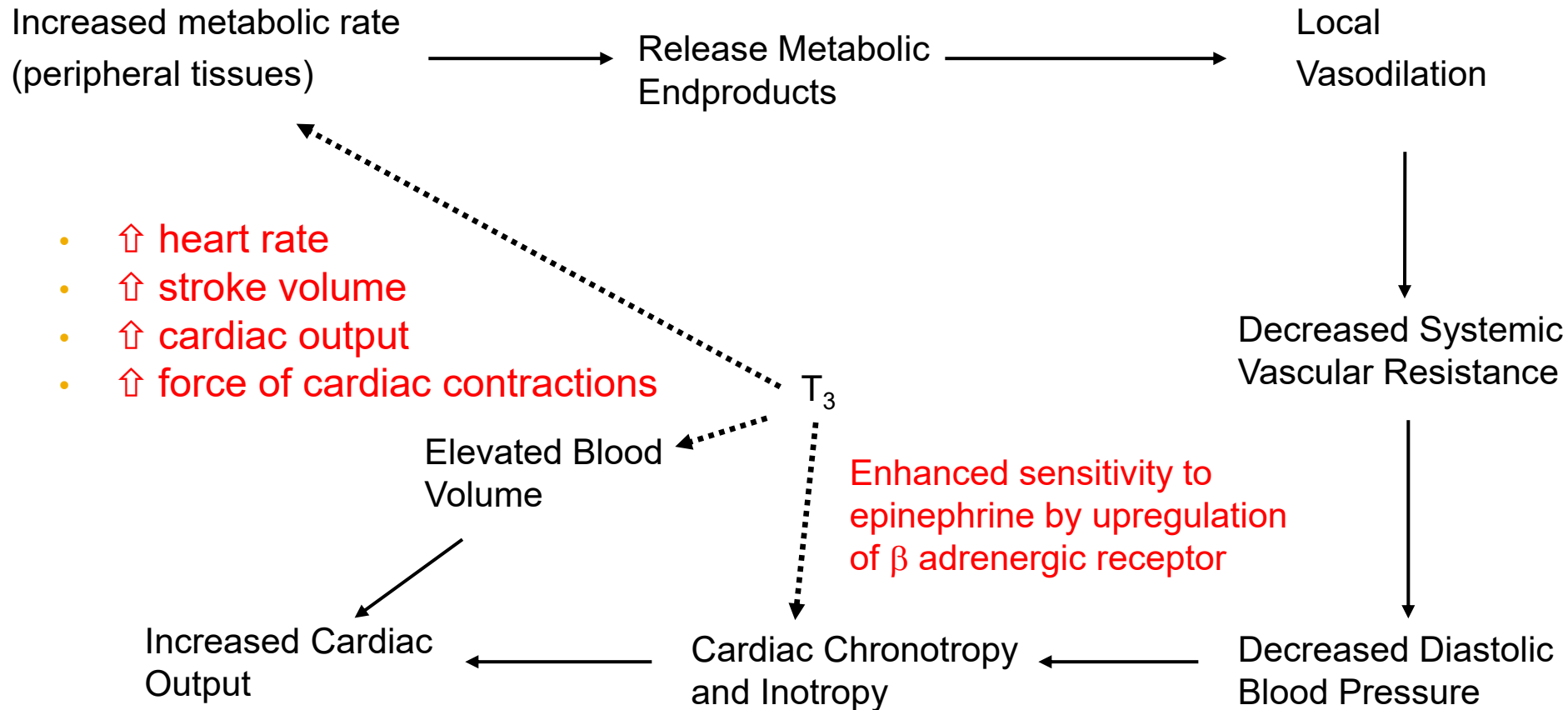


Essential for CNS development

- promotes nerve growth
- growth of dendrite cells
- myelin formation

Maternal and/or fetal hypothyroidism causes delayed CNS development and intellectual disability

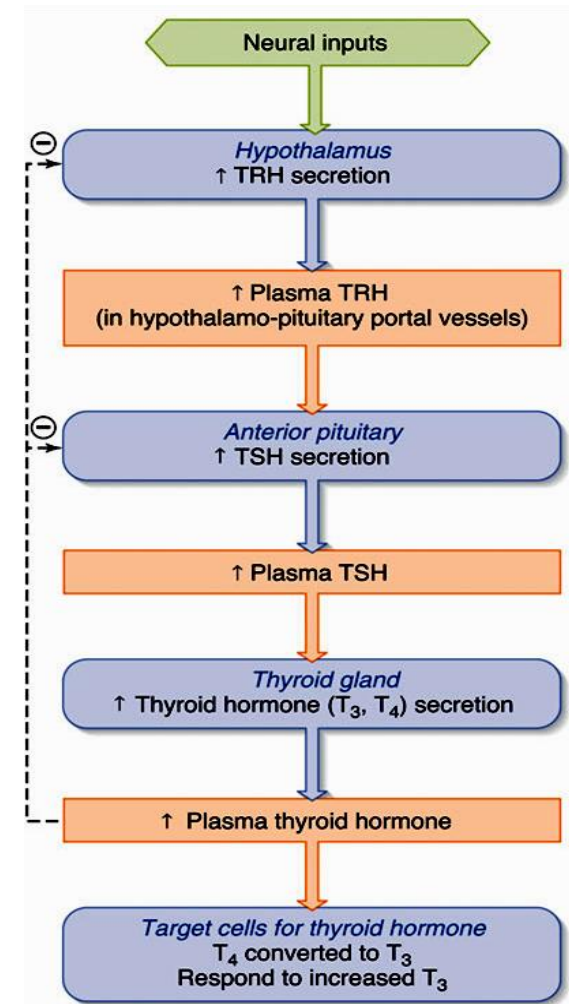
TH Influences on Cardiovascular Hemodynamics



Changes in the cardiovascular system are prominent clinical consequences in thyroid dysfunction

Regulation of Thyroid Function: Hypothalamus-Pituitary-Thyroid Axis

- Anterior pituitary
 - Thyroid stimulating hormone (TSH)
 - $\uparrow$ TH synthesis
 - stimulates follicular cell growth, follicle formation
 - maintenance of function and **growth of gland**
- Hypothalamus
 - TRH regulates secretion of pituitary TSH
 - Hypothermia is major stimulus to TRH secretion
- Feedback regulation
 - Maintains relatively stable T3/T4 output by thyroid (compared with other hormones)



Thyroid Dysfunction

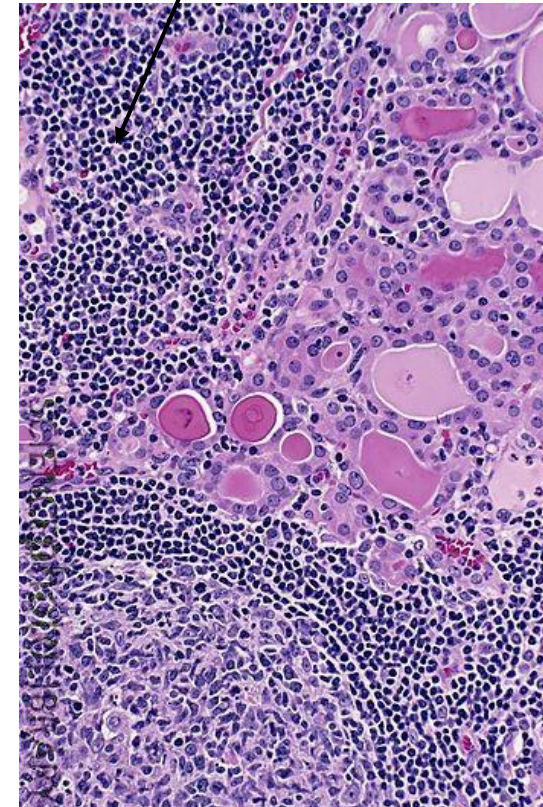
- Hypothyroidism
 - T3/T4 insufficiency
- Hyperthyroidism
 - T3/T4 excess
- Goitre, enlargement of thyroid gland
 - diffuse
 - nodular (uni or multinodular)
- Caused by thyroid overstimulation by TSH
- Observed in both hypo- and hyperthyroid states



Hypothyroidism : Causes

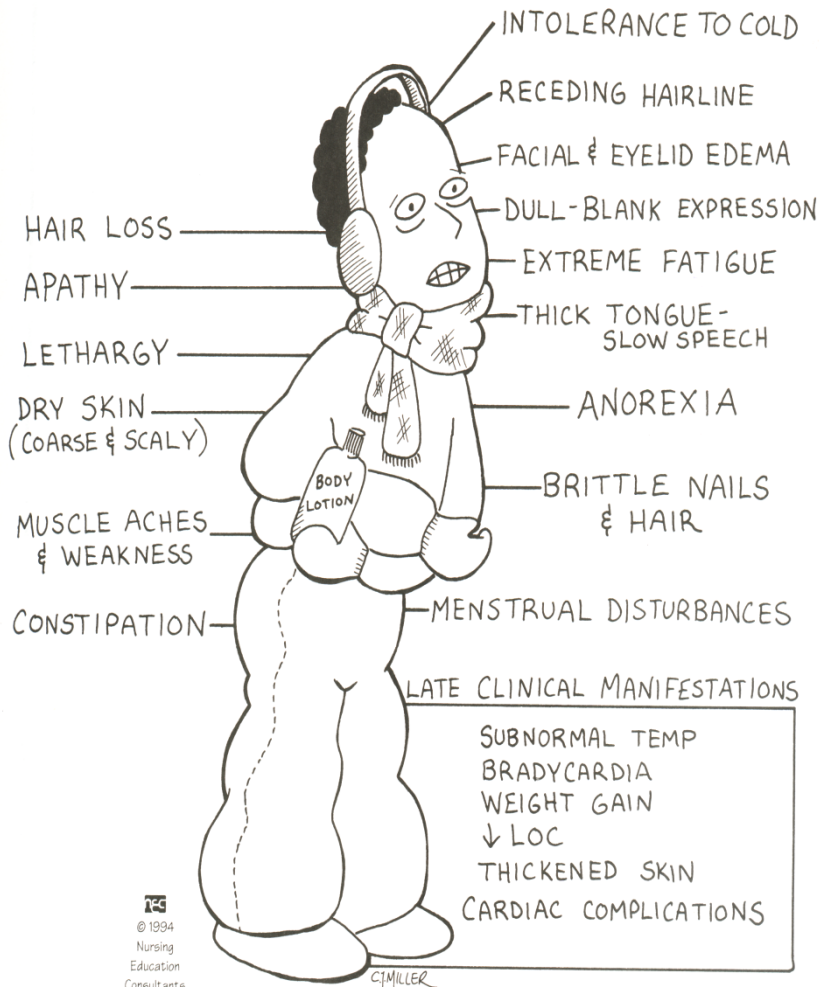
- Primary hypothyroidism (abnormality of thyroid itself)
 - iodine deficiency
 - genetic (mutations in receptors or enzymes)
 - toxicity (eg. thiocyanate, perchlorate)
 - other disease processes (eg. cancer)
 - autoimmune – Hashimotos Thyroiditis
 - autoantibodies to TSH receptor which act as **TSH antagonists**
 - other autoantibodies to thyroid peroxidase/thyroglobulin
- Secondary hypothyroidism (abnormality of hypothalamus or pituitary)

infiltrating immune cells



Hypothyroidism : Symptoms#

HYPOTHYROIDISM



• Adults

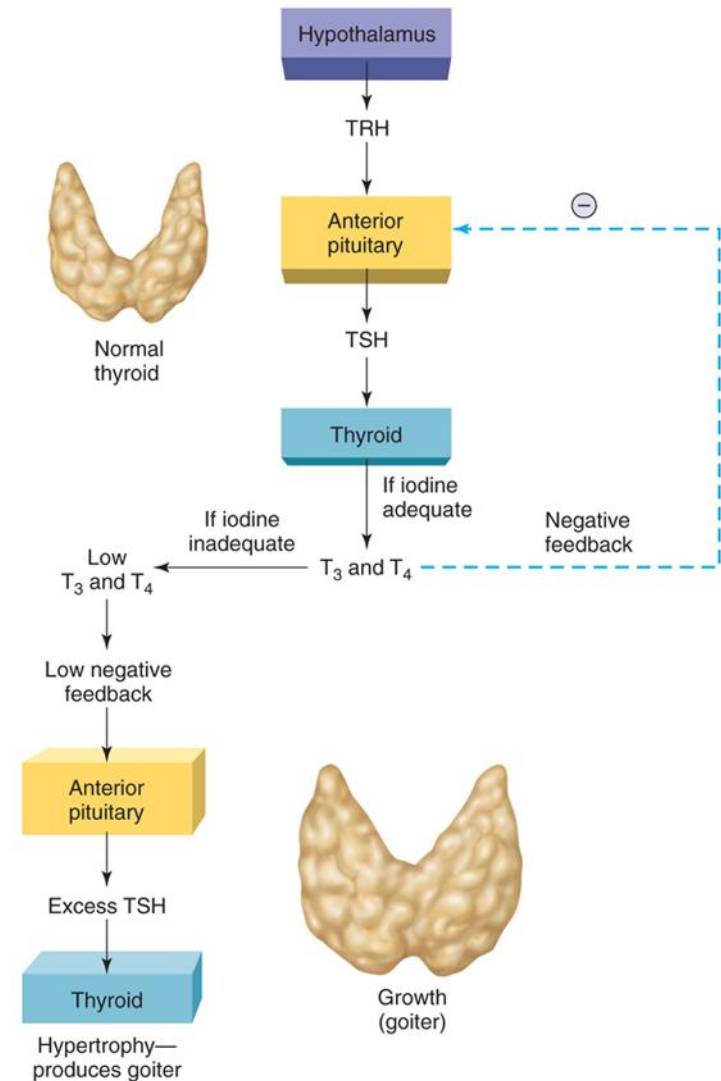
- Goitre
- Decreased heart rate and cardiac output
- Decreased appetite
- Lethargy/somnolence
- Decreased basal metabolism
- Cold intolerance
- Psychological disorders (depression, anxiety)

• Children

- Intellectual disability, dwarfism (cretinism)

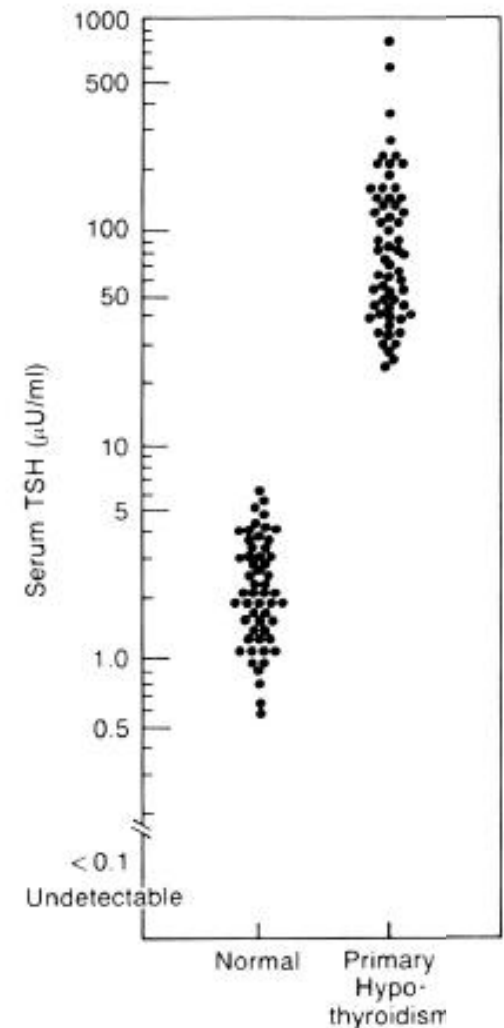
Goitre in Hypothyroid States

- Low levels of TH production leads to reduced negative feedback
 - $\uparrow$ TSH production
 - generalised hypertrophy of gland



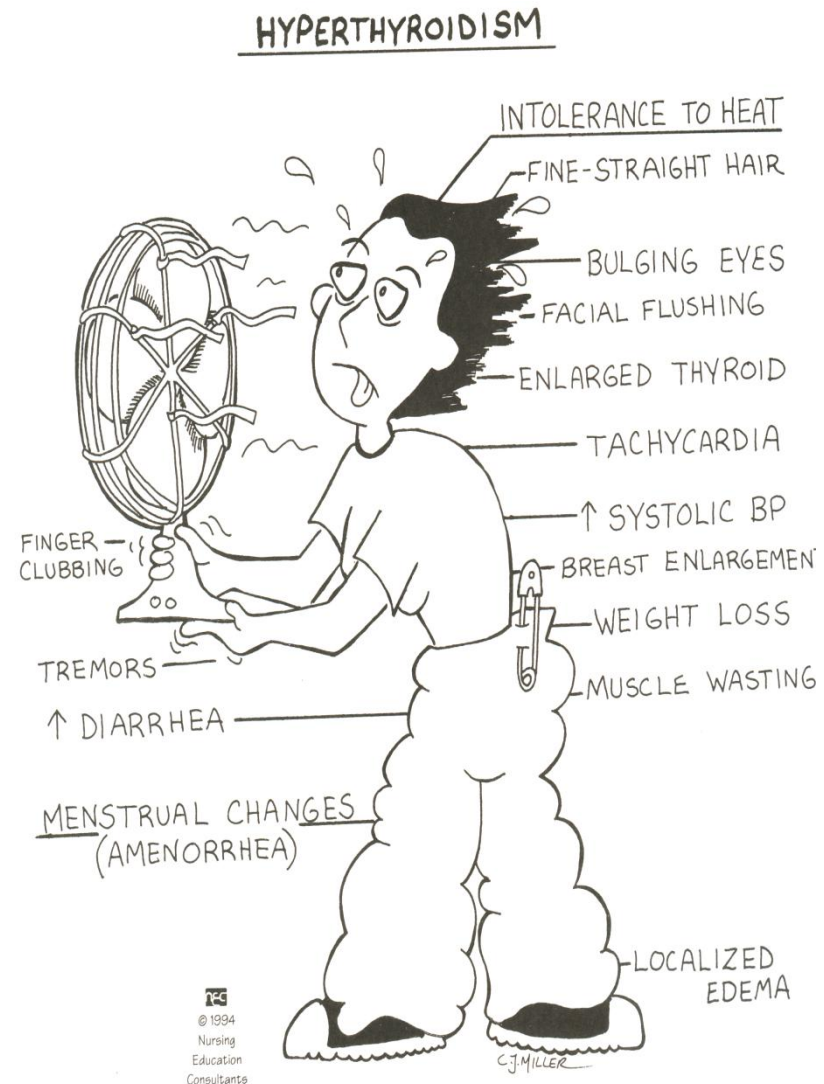
Primary Hypothyroidism : Diagnosis and Treatment

- Diagnosis
 - decreased T_3/T_4 production
 - increased TSH
 - presence of anti-thyroid antibody (autoimmune etiology)
- Treatment
 - replacement therapy with **levothyroxine**
 - dietary iodine supplementation
- Secondary hypothyroidism
 - decreased T_3/T_4 production
 - **decreased** TSH



Hyperthyroidism : Symptoms

- **Thyrotoxicosis**
- 10 times more common in women
- Increased basal metabolic rate
- Heat intolerance
- Increased body temperature
- Increased heart rate / arrhythmias
- Nervousness, anxiety
- Goitre
- Exophthalmos ('bulging eyes')



Graves Exophthalmos

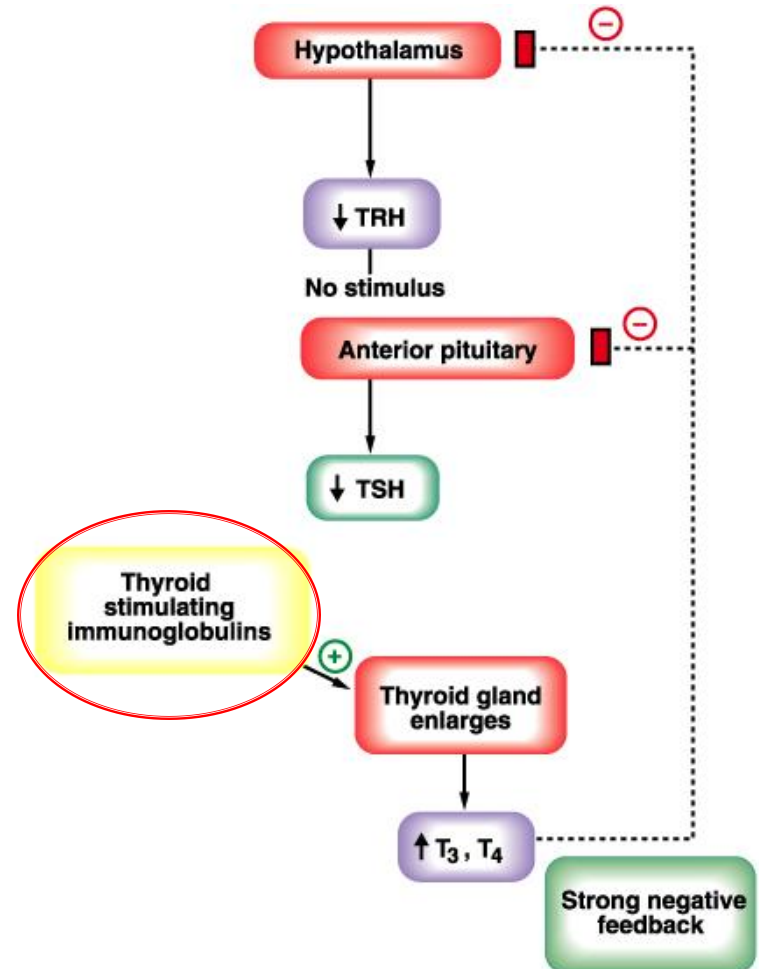


Hyperthyroidism : Causes

- Primary hyperthyroidism (abnormality of thyroid itself)
 - autoimmune disorder – Graves Disease
 - autoantibodies to TSH receptor which act as TSH agonists
 - T3/T4 hypersecretion
 - stimulus cannot be overridden by feedback regulation
 - thyroid adenoma
- Secondary hyperthyroidism (abnormality of hypothalamus or pituitary)
 - excess TSH (pituitary) – TSH secreting adenoma
 - excess TRH (hypothalamus)

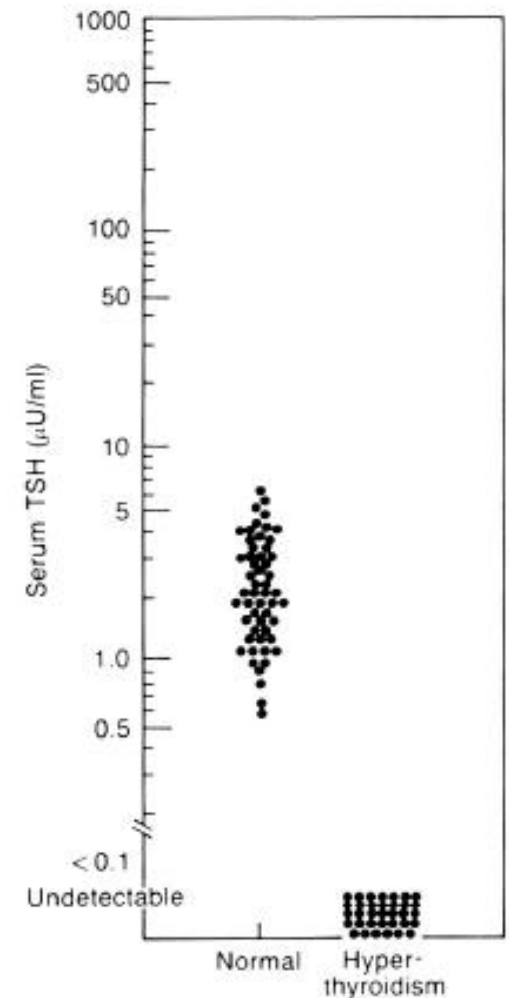
Goitre in Graves Disease

- Autoimmune disorder
- Autoantibodies directed against TSH receptor
 - stimulating antibodies
 - mimic effect of TSH binding
 - stimulates TH production and hypertrophy of thyroid gland
- Increased feedback inhibition of pituitary TSH production
 - ineffective because TSH is not cause of thyroid stimulation



Primary Hyperthyroidism : Diagnosis and Treatment

- Diagnosis
 - increased T_3/T_4 production
 - decreased TSH
 - presence of anti-TSH stimulatory antibodies (autoimmune)
- Treatment
 - agents that interfere with production of T_3 and T_4
 - thyrostatic drugs (carbimazole)
 - destruction of thyroid gland
 - thyroidectomy
 - radioablation of follicular
- Secondary hyperthyroidism
 - increased T_3/T_4 production
 - **increased** TSH



Thyrotoxic Crisis (Thyroid Storm)

- Thyroid Storm
 - extreme hyperthyroidism
 - hypermetabolic state characterised by mortality rate high (10-50%)
- Symptoms
 - tachycardia (>140bpm)
 - cardiac arrhythmias
 - congestive heart failure
 - pyrexia (>38.5°C)
 - dehydration
- Treatment
 - β blockers (reduce sympathetic activation)
 - glucocorticoids (reduce T4 to T3 conversion)
 - thionamides (inhibit new hormone synthesis)

Drug	Action
Antithyroid agents propylthiouracil (also called PTU) methimazole (Tapazole)	Prevents production of more T ₄ and T ₃ in the thyroid, and blocks the conversion of T ₄ to T ₃ outside the thyroid. Prevents production of more thyroid hormone.
Iodides Lugol's solution saturated solution of potassium iodide (Pima, SSKI)	Blocks release of stored thyroid hormone from thyroid gland.
Glucocorticoids dexamethasone (Decadron) hydrocortisone	Blocks conversion of T ₄ to T ₃ .
Beta-blockers propranolol (Inderal) esmolol (Brevibloc Injection)	Reduces symptoms (tachycardia, tremor, restlessness) caused by a heightened response to catecholamines; blocks conversion of T ₄ to T ₃ .

Next

Fri Oct 2nd @ 12:00 in C005
Calcium Regulation

Wednesday Oct 7th @ 17:00 in B004
Thyroid / Parathyroid Tutorial